Vehicle Price Prediction Using Extremely Randomized Trees Regression

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Introduction

The price of a vehicle listed in an online marketplace is influenced by multiple attributes, including age, model, mileage, fuel type, transmission, and body style. Buyers and sellers often use these characteristics when estimating vehicle value, while analysts use them to study pricing trends and predict market behavior. Machine learning techniques can help identify which variables have the greatest influence on price and can improve predictive accuracy compared to simple descriptive comparisons.

This project explores vehicle pricing using regression modeling in Python. The purpose of this study was to determine which vehicle characteristics most strongly predict advertised vehicle price among a selected subset of car advertisements. Python was used for data cleaning, exploratory data analysis, visualization, encoding of categorical variables, and predictive modeling.

Objective

The objective of this analysis was to determine the most influential attributes when predicting advertised vehicle price among selected vehicle advertisements. The project specifically examined the following vehicle models:

- Mitsubishi L200
- Audi Q3
- Mazda CX-5
- Volvo XC90

The analysis was limited to sport utility vehicles and pickup trucks using petrol or diesel fuel and the six most frequently advertised vehicle colors.

Dataset and Sample

The sample data used in this research was adapted from car advertisement data collected by Huang et al. (2021). The sample represents online vehicle advertisements for Mitsubishi L200, Audi Q3, Mazda CX-5, and Volvo XC90 vehicles. The advertisements included sport utility vehicles and pickup trucks using petrol or diesel fuel and the six most frequently advertised vehicle colors.

The information analyzed included vehicle model, registration year, color, body type, mileage, gearbox type, fuel type, and advertised price. After applying the specified filtering criteria and removing incomplete observations, the final analysis dataset contained 3,270 vehicle advertisements.

Methodology

The data were analyzed using Extremely Randomized Trees Regression to determine the influence vehicle characteristics have when predicting advertised vehicle price. Extremely Randomized Trees Regression is an ensemble machine learning method that evaluates multiple randomized decision trees and combines their predictions to improve overall model performance.

One weakness of tree-based machine learning models is the potential for overfitting, where a model learns patterns specific to the training data rather than generalizable relationships. To mitigate this weakness, the data were separated into training and testing datasets. Model performance was evaluated on both datasets to determine whether the model generalized well to unseen observations.

Another consideration involves feature importance measures. Traditional tree-based importance metrics may be biased toward variables with specific characteristics. To address this weakness, permutation importance was used to evaluate the influence of predictor variables on model performance (Parr et al., 2018). This approach provides a more reliable assessment of attribute importance and improves the validity of the conclusions.

The workflow included data preparation, exploratory analysis, model training, model validation, and feature importance assessment.

Data Preparation

Data preparation included:

- importing the advertisement dataset,
- restricting the data to the required vehicle models,
- limiting observations to SUV and pickup body styles,
- limiting observations to petrol and diesel fuel types,
- retaining the six most common vehicle colors,
- removing incomplete observations.

Exploratory Data Analysis

Exploratory analysis included visual inspection of vehicle price distributions and comparisons of advertised prices across vehicle models.

Figure 1 presents the distribution of vehicle prices in the cleaned dataset.

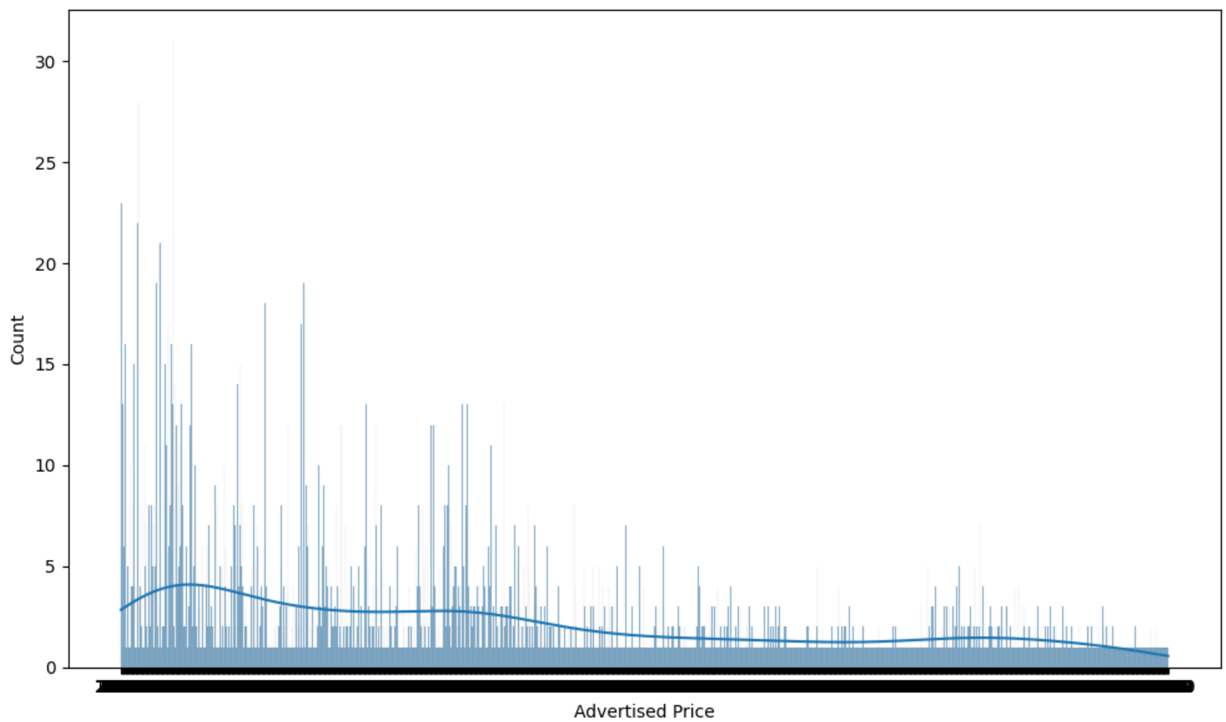

Figure 1. Distribution of advertised vehicle prices in the cleaned dataset.

Visualizations included:

- histogram of vehicle price distribution,
- boxplot of vehicle price by model,
- feature importance bar chart.

Model Training

The dataset was split into:

- 80% training data
- 20% testing data

The model was evaluated using:

- coefficient of determination (R^2)
- root mean squared error (RMSE)

Permutation importance was then used to assess the relative influence of each predictor variable.

Results

The regression model demonstrated strong predictive performance.

Figure 2 presents the distribution of vehicle prices among the four vehicle models included in the analysis.

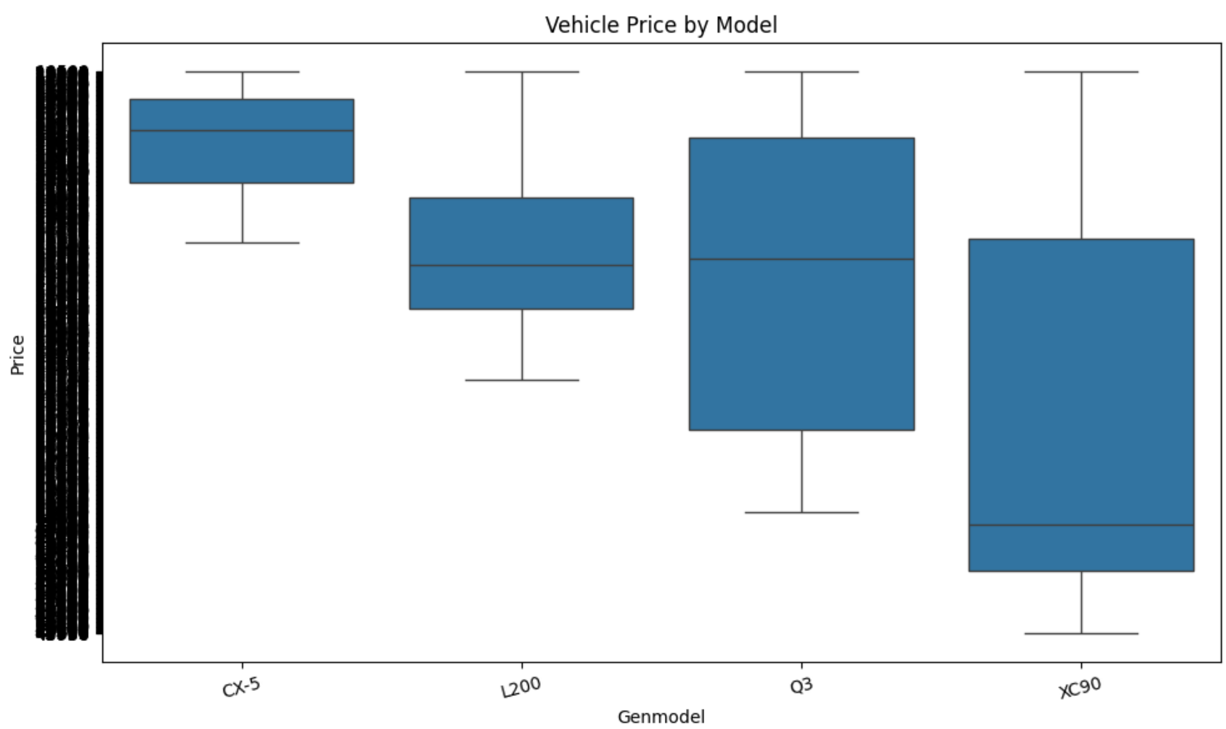

Figure 2. Distribution of vehicle prices across the Mitsubishi L200, Audi Q3, Mazda CX-5, and Volvo XC90 models.

Model Performance

Metric	Training	Testing
R^2	0.9991	0.9075
RMSE	300.17	2881.55

The data were analyzed with Extremely Randomized Trees Regression to determine the influence vehicle characteristics have when predicting advertised price. Model validity was assessed through training and testing. The training model produced an R^2 value of .9991 and an RMSE of 300.17. The testing model produced an R^2 value of .9075 and an RMSE of 2881.55.

The similar performance between the training and testing datasets suggests that the model is valid and reasonably reliable for predicting advertised vehicle price. Although testing performance was lower than training performance, the model continued to explain over 90% of the variation in vehicle prices.

Feature Importance

Figure 3 presents the permutation importance scores generated by the regression model.

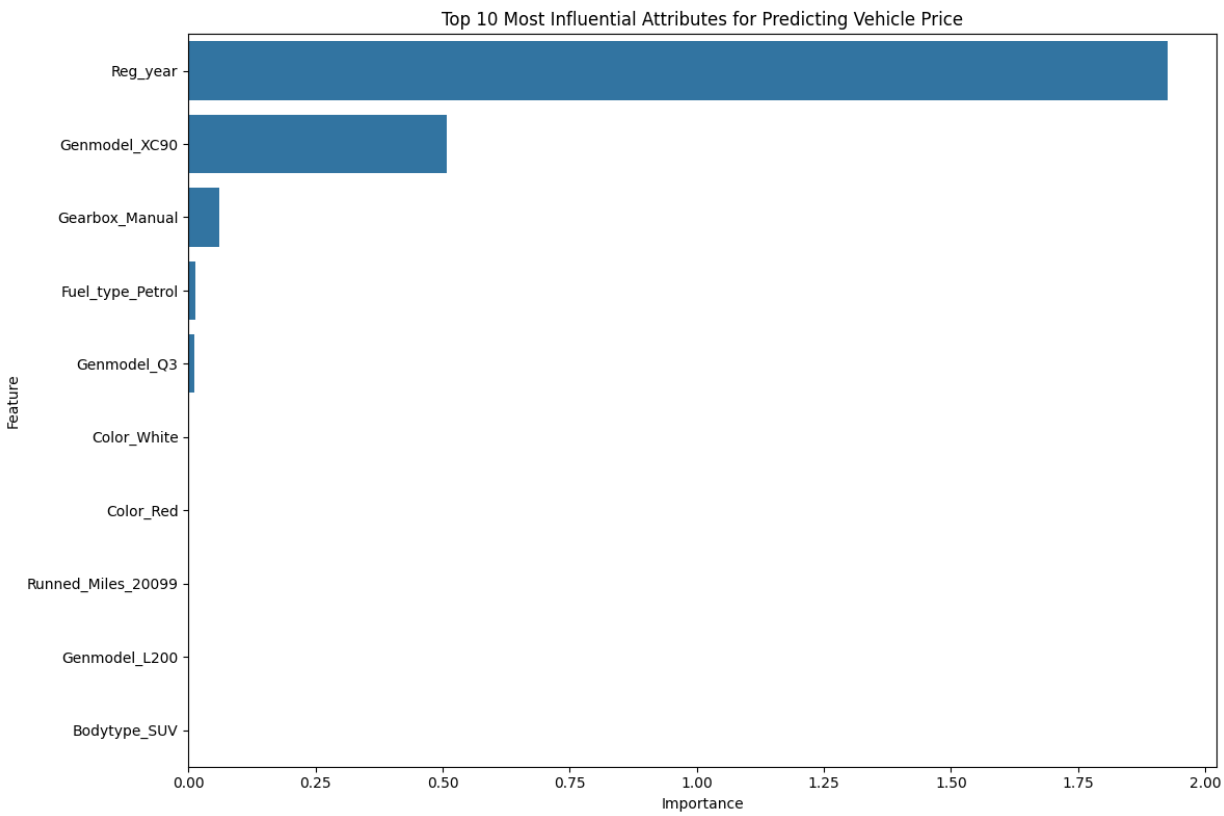

Figure 3. Permutation feature importance scores generated by the Extremely Randomized Trees Regression model.

The most influential predictors of vehicle price were:

1. Registration Year (Reg_year)
2. Volvo XC90 model
3. Manual gearbox
4. Petrol fuel type
5. Audi Q3 model

Registration year was by far the strongest predictor of vehicle price.

Discussion

When the model was assessed for permutation-based importance, registration year was the most influential attribute when predicting vehicle price. Only a small number of additional variables demonstrated meaningful influence, including vehicle model and gearbox type.

The permutation importance measures suggest that registration year, vehicle model, and gearbox type were the characteristics that were most influential when predicting an accurate advertised vehicle price. These findings directly address the research objective by identifying the vehicle attributes with the greatest influence on price prediction.

The results suggest that vehicle age is the strongest factor affecting vehicle value within this sample. Newer vehicles generally command higher advertised prices, which aligns with expected trends in automotive resale markets. Vehicle model was also important, particularly the Volvo XC90, suggesting that manufacturer reputation and vehicle positioning contribute significantly to market value.

Gearbox type and fuel type demonstrated smaller but measurable influence. These variables likely reflect buyer preferences and demand characteristics within specific vehicle segments. Mileage contributed some predictive value, although its influence was lower than expected compared to registration year.

Color demonstrated minimal influence on pricing. While vehicle color may affect consumer preference, the results indicate that structural vehicle characteristics have substantially greater influence on advertised price.

One limitation of this research is that advertised prices were analyzed rather than completed transaction prices. Advertised prices may differ from final sales values. Additionally, other potentially important variables such as vehicle condition, service history, accident history, and geographic market conditions were not available in the dataset.

Conclusion

This research examined vehicle advertisements to determine the most influential attributes when predicting vehicle price. Using Extremely Randomized Trees Regression, the analysis identified registration year as the strongest predictor of vehicle price, followed by vehicle model and gearbox type.

The model demonstrated strong predictive performance and maintained similar performance between training and testing datasets, supporting the validity and reliability of the findings.

The results indicate that vehicle age, model, and transmission characteristics are the most influential attributes when predicting advertised vehicle price within this sample. These findings demonstrate how machine learning methods can be used to identify important predictors in large advertisement datasets and support data-driven decision making in vehicle valuation analysis.

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